AI Product Design

Knowledge Graphs vs Context Graphs

1. Knowledge Graphs

Definition

Structured representation of entities (people, places, concepts) and their relationships.

Purpose

Captures what exists and how things connect.

Examples in AI Products

- Recommendation engines linking users → products → categories → reviews.
- Medical AI linking symptoms → diseases → treatments.

Knowledge Graphs: Analysis

Strengths

+ Clear, factual relationships.
+ Easy to query and visualize.
+ Supports reasoning and inference.

Limitations

– Static: doesn't adapt well to changing situations.
– May miss contextual meaning (e.g., "bank" as riverbank vs financial institution).

2. Context Graphs

Definition

Dynamic representation of situational or conversational context around knowledge.

Purpose

Captures when, why, and under what conditions relationships matter.

Examples in AI Products

- Chatbots adapting answers based on user's intent, tone, or history.
- Travel AI adjusting recommendations based on season, budget, or group size.

Context Graphs: Analysis

Strengths

+ Flexible, adaptive to scenarios.
+ Adds nuance to knowledge graphs.
+ Enables personalization and situational awareness.

Limitations

– More complex to design and maintain.
– Requires continuous updates and contextual signals (time, location, user state).

3. Ease vs Complexity

Aspect	Knowledge Graphs	Context Graphs
Ease of Use	Easier to build, query, and visualize	Harder—requires dynamic inputs and contextual modeling
Complexity	Lower—mostly static relationships	Higher—dynamic, multi-layered context
Best Use Case	Structured data, factual reasoning	Adaptive AI, personalization, scenario-driven design
Challenge	Ambiguity in meaning	Managing evolving context signals

4. Role in AI Product Design

Knowledge Graphs

Provide the foundation—the "map" of entities and relationships.

Context Graphs

Provide the lens—adjusting the map to the situation at hand.

Together

They enable AI systems that are both accurate (knowledge) and relevant (context).

Summary

Think of Knowledge Graphs as the skeleton (structure) and Context Graphs as the muscles and nerves (adaptation). One gives stability, the other gives responsiveness.

CASE STUDY

Real-World Scenario

Email Prioritization by Behavior

Knowledge Graph Approach

What it captures

Entities and relationships in the email ecosystem.

Structure

- Entities: User, Sender, Email, Topic, Attachment, Time.
- Relationships: User → frequently replies to → Sender A
- Email → contains → Topic X
- Sender B → belongs to → Project Y

Usage

- The system can infer that emails from Sender A or about Project Y are more important.
- Prioritization is based on static relationships (frequency, project membership, topic relevance).

Analogy

A map of who's connected to whom and what topics matter.

Context Graph Approach

What it captures

Situational and behavioral signals around those relationships.

Contextual Signals

- Time of day: User tends to reply to work emails in the morning.
- Device: User ignores long attachments when on mobile.
- Recent activity: User just searched for "budget report," so emails with "budget" gain priority.
- Emotional tone: User responds faster to concise, action-oriented emails.

Usage

- Morning → prioritize work emails.
- Evening → prioritize personal/family emails.
- After a project deadline → deprioritize project-related threads.

Analogy

A lens that changes the map depending on the situation.

Classroom Exercise

Step 1

Students sketch a knowledge graph of email relationships (user ↔ sender ↔ topic).

Step 2

Extend it into a context graph by adding behavioral signals (time, device, recent searches).

Step 3

Discuss how prioritization changes when context shifts (e.g., morning vs evening, project vs vacation).

Key Takeaway

Knowledge Graph = "Who/What matters."

Context Graph = "When/Why it matters."